

- Mean function

$$\mu(t) = E[X_t]$$

- Covariance (autocovariance) function

$$\gamma(s, t) = \text{Cov}(X_s, X_t)$$

$$= E[(X_s - E[X_s]) \cdot (X_t - E[X_t])]$$

- Useful fact:

$\text{Cov}(X, Y)$ behaves like product
 $X \cdot Y$.

E.g. $X_t = \frac{1}{3}(w_{t-1} + w_t + w_{t+1})$

where w_t : white noise, $\text{var}(w_t) = \sigma_w^2$.

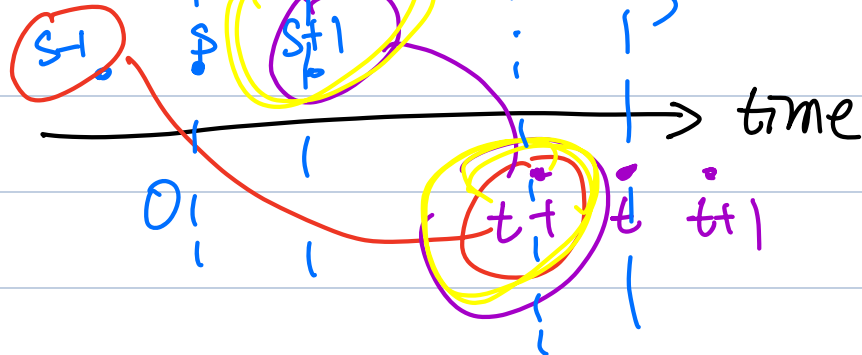
Goal: compute $\gamma(s, t)$.

$$\gamma(s, t) = \text{Cov}(X_s, X_t)$$

$$= \text{Cov}\left(\frac{1}{3}(w_{s-1} + w_s + w_{s+1}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1})\right)$$

$$\rightarrow = \left(\frac{1}{3} \times \frac{1}{3}\right) \text{Cov}(w_{s-1} + w_s + w_{s+1}, w_{t-1} + w_t + w_{t+1})$$

$i \quad j \quad k$
 $1 \quad 2 \quad 3$



Case 1: $|s-t| \geq 3$.

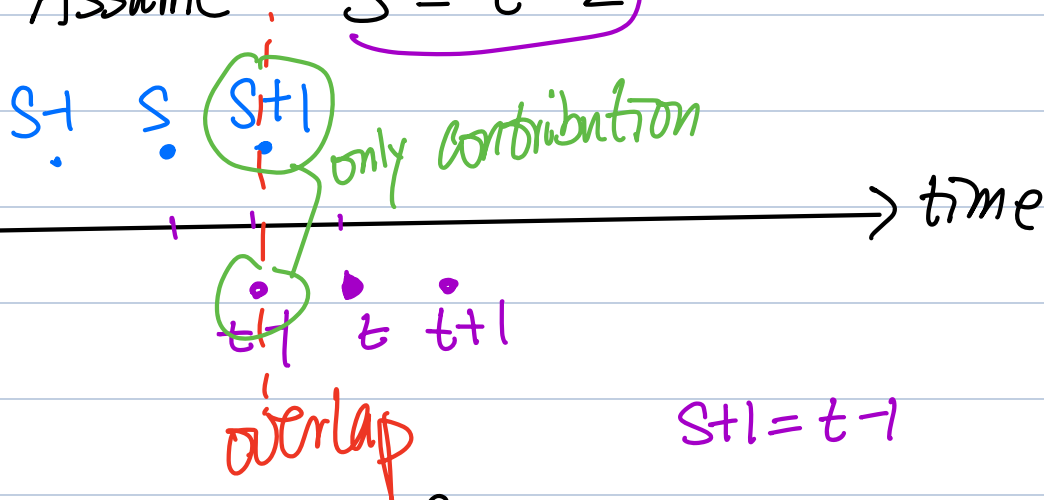
$$\text{Cov}(W_j, W_k) = 0$$

for all $j = s-1, s, s+1$, & $k = t-1, t, t+1$.

$$\Rightarrow \gamma(s, t) = 0$$

Case 2: $|s-t| = 2$

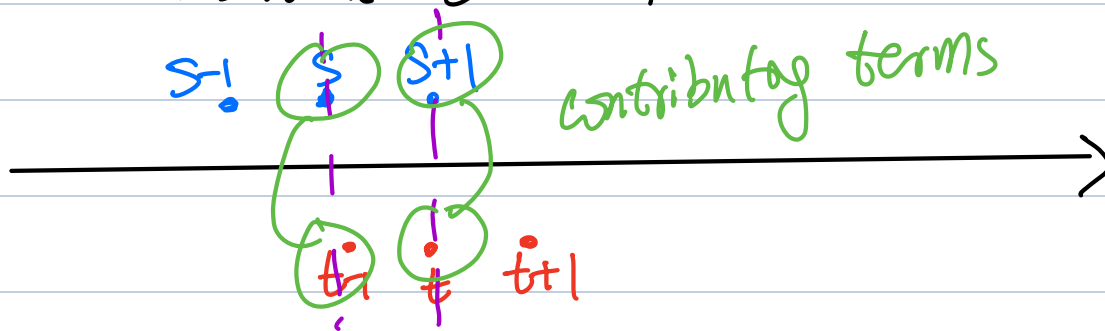
Assume $S = t-2$



$$\begin{aligned} \gamma(s, t) &= \frac{1}{3} \times \frac{1}{3} \text{Cov}(W_{s+1}, W_{t-1}) \\ &= \frac{1}{9} \sigma_w^2 \end{aligned}$$

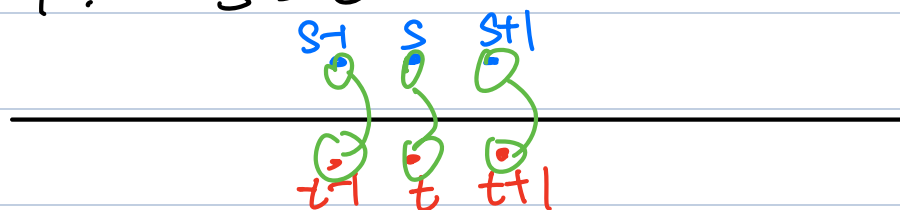
Case 3: $|s-t| = 1$

Assume $s = t-1$



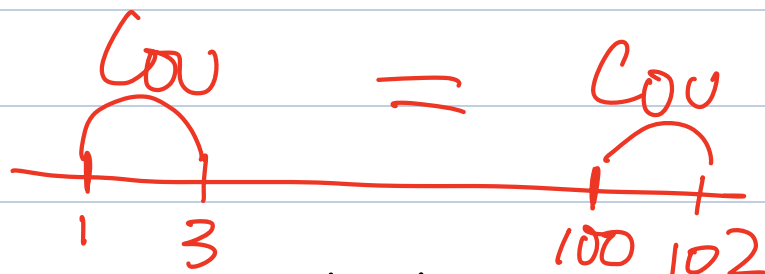
$$\gamma(s, t) = \dots = \frac{2}{q} \sigma_w^2$$

Case 4: $s = t$



$$\gamma(s, t) = \frac{3}{q} \sigma_w^2$$

To sum up:



$$|s-t| \geq 3$$

$$|s-t| = 2$$

$$|s-t| = 1$$

$$s = t$$

$$|s-t| \geq 3$$

$$|s-t| = 2$$

$$|s-t| = 1$$

$$s = t$$

$$\rho(s, t) = \frac{\gamma(s, t)}{\sqrt{\gamma(s, s)\gamma(t, t)}} =$$

E.g.

$$\begin{cases} X_t = X_{t-1} + w_t \\ X_0 = 0 \end{cases}$$

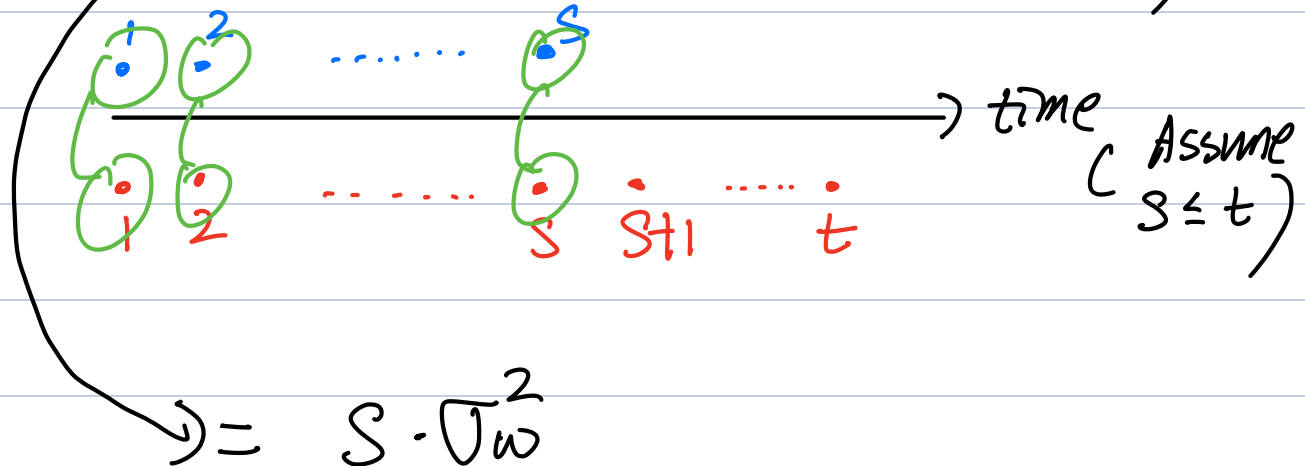
0.1111

w_t : white noise
 σ_w^2

$$\Rightarrow X_t = \omega_1 + \omega_2 + \dots + \omega_t$$

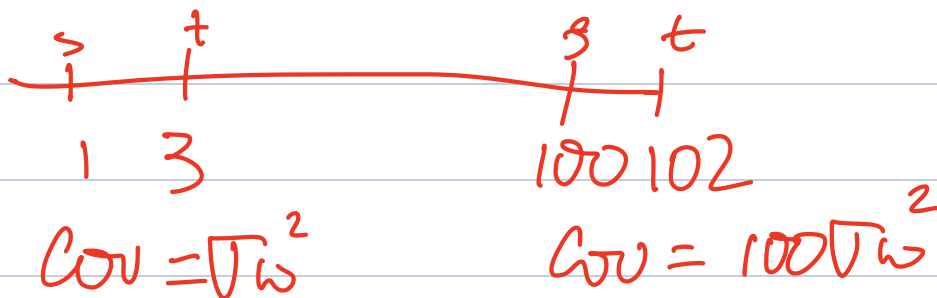
$$\gamma(s, t) = \text{Cov}(X_s, X_t)$$

$$= \text{Cov}(\omega_1 + \omega_2 + \dots + \omega_s, \omega_1 + \dots + \omega_t)$$



Similarly, if $s > t$,
 $\gamma(s, t) = t \cdot \sigma_{\omega}^2$.

$$\boxed{\gamma(s, t) = \sigma_{\omega}^2 \min(s, t)}$$



Def. $\{x_t\}$: a time series

$\gamma(s, t)$: cov. function

Define correlation (autocorrelation)

function as:

ACF

$$\begin{aligned} \rho(s, t) &= \text{Corr}(X_s, X_t) \\ &= \frac{\text{Cov}(X_s, X_t)}{\sqrt{\text{Var}(X_s)} \cdot \sqrt{\text{Var}(X_t)}} \\ &= \frac{\gamma(s, t)}{\sqrt{\gamma(s, s) \gamma(t, t)}} \end{aligned}$$

Remark: $\rho(s, t) \in [-1, 1]$

Exercise 11: Derive ACF for
signal + noise model.

Hints: $\text{Corr}(aX + b, cY + d)$

$$= \begin{cases} \text{Corr}(X, Y) & \text{if } a \cdot c > 0 \\ -\text{Corr}(X, Y) & \text{if } a \cdot c < 0 \end{cases}$$

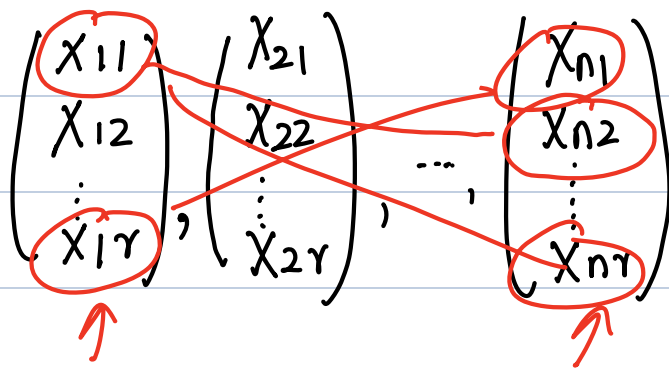
where: X & Y are RVs.

a, b, c, d are non-random constants

$a, c \neq 0$.

- Now consider an r -dimensional multivariate time series:

$$\vec{X}_1, \vec{X}_2, \dots, \vec{X}_n$$

Def. The mean function of $\{\vec{X}_t\}$

$$\vec{\mu}(t) = \begin{pmatrix} E[X_{t1}] \\ E[X_{t2}] \\ \vdots \\ E[X_{tr}] \end{pmatrix}$$

The (cross)-covariance function
(CCF)
is defined:

$$\gamma_{jk}(s, t) = \text{Cov}(X_{sj}, X_{tk})$$

The (cross)-correlation function
is defined as:

$$\rho_{jk}(s, t) = \text{Corr}(X_{sj}, X_{tk})$$

Rem: when $j=k$,

$\gamma_{jj}(s, t), \rho_{jj}(s, t)$ are cov/corr. function

for the univariate time series at j^{th} component.

E.g. $\{w_t\}$: white noise, σ_w^2 .

Define:
$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} w_t + w_{t-1} \\ w_t - w_{t-1} \end{pmatrix}$$

Goal: compute mean & cov. function.

$$\begin{aligned} \vec{\mu}(t) &= \begin{pmatrix} E[x_t] \\ E[y_t] \end{pmatrix} = \begin{pmatrix} E[w_t + w_{t-1}] \\ E[w_t - w_{t-1}] \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{aligned}$$

$$\gamma_{xx}(s, t) = \text{Cov}[X_s, X_t] = \dots = \begin{cases} 2\sigma_w^2 & |s-t|=0 \\ \sigma_w^2 & |s-t|=1 \\ 0 & |s-t| \geq 2 \end{cases}$$

$$\gamma_{yy}(s, t) = \text{Cov}(Y_s, Y_t)$$

Case 1: $|s-t| \geq 2$

$$\gamma_{yy}(s, t) = 0$$

Case 2: $|s-t|=1$

Assume $s = t-1$

$s-1$ s

time

$t-1$ t

$$\gamma(s, t) = \text{Cov}(W_s - W_{s-1}, W_t - W_{t-1})$$

$$= \text{Cov}(W_s, -W_{t-1}) + 0$$

$$= -\sigma_w^2$$

Case 3: $s=t$

$$\gamma(s, s) = \text{Cov}(W_s - W_{s-1}, W_s - W_{s-1})$$

$$= \text{Cov}(W_s, W_s) + \text{Cov}(-W_{s-1}, -W_{s-1}) + 0$$

$$= \sigma_w^2 + (-1) \cdot (-1) \sigma_w^2$$

$$= 2\sigma_w^2$$

$$\gamma_{xy}(s, t) = \text{Cov}(X_s, Y_t)$$

details

Optional
Exercise

$$\begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix} = \begin{cases} 0 \\ \sigma_{\omega}^2 \\ -\sigma_{\omega}^2 \\ 0 \end{cases}$$

$$|s-t|=0$$

$$s-t=1$$

$$s-t=1$$

$$|s-t| \geq 2$$

★ Remark: $\gamma_{xy}(s, t)$ may differ from $\gamma_{xy}(t, s)$
vs.

For univariate case

$$\gamma(s, t) = \gamma(t, s)$$